

IDEA for Operators: Tutorial

The Console, disks, backups and classroom computer cards

Version 1.1, 22 October 2026

Contents

1. Introduction 3
2. The Console for the operator 4
  2.1 Operator Login 4
  2.2 The Network view: the IDEA box and its disks 5
  2.3 App status 5
  2.4 Command History 6
  2.5 What the buttons do 6
3. Disks: Files Disk and Backup Disk 7
  3.1 A new disk (not yet an IDEA Disk) 7
  3.2 Erase a disk 7
  3.3 Make a Files Disk 8
  3.4 Eject a disk 9
  3.5 Make a Backup Disk 10
  3.6 Run a backup 12
  3.7 Restore after a breakdown 12
4. Student computers: reset a card from the Console 16
  4.1 Before you start 16
  4.2 Find the card 16
  4.3 Reset (flash) the card 17
  4.4 After the reset 19
  4.5 Questions 20

1. Introduction

The **operator** is the person at the school who looks after the IDEA box: the small computer that runs Kolibri, Nextcloud and Wikipedia for the whole school, without internet. This tutorial shows the operator, step by step, how to sign in to the Console, check that the apps run, look after the disks and the backups, and reset a classroom computer's card. None of it needs internet, and none of it needs a technician: when something is beyond these steps, call your IDEA contact.

The teachers' side (opening apps, Kolibri, Nextcloud, Wikipedia) is in IDEA for Teachers: Tutorial. The words used here (IDEA box, Console, IDEA Disk, Files Disk, Backup Disk) are explained in its chapter 1.

2. The Console for the operator

2.1 Operator Login

1. In the Console, click the person icon (top right). The **Operator Login** panel opens with a username and a password field (the eye icon shows or hides the password).

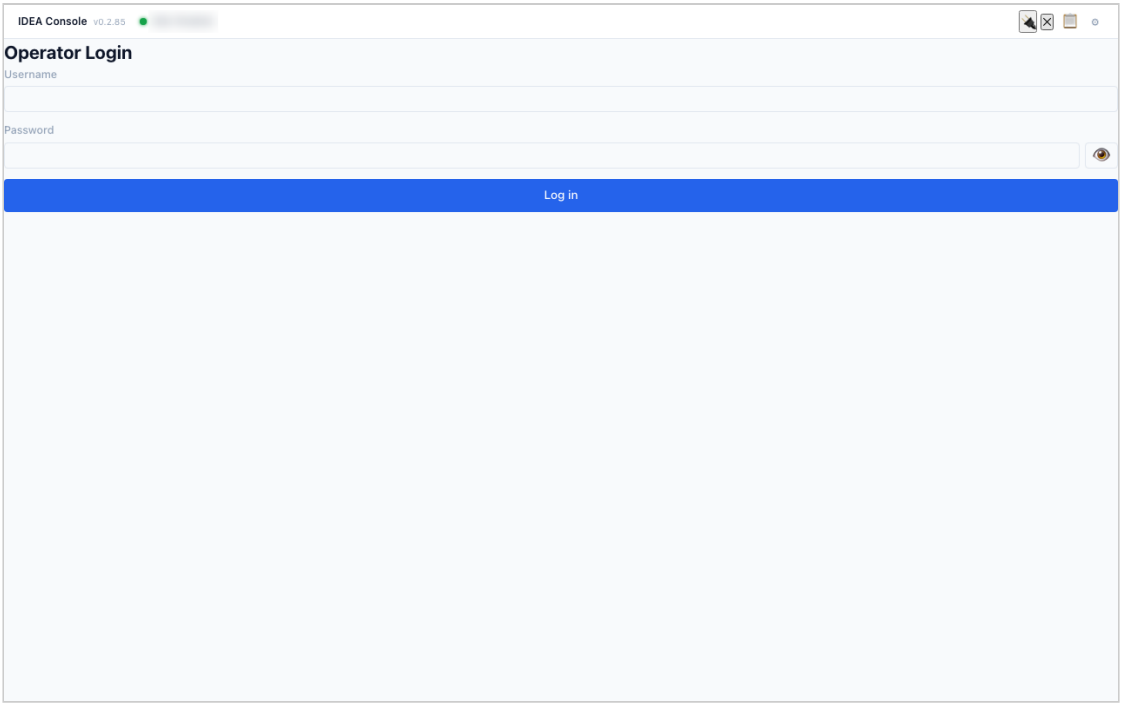


Figure 1. Operator Login

2. Type the operator username and password (they are on the IDEA Kit Accounts and Passwords sheet) and click **Log in**. The **Account** panel shows your name, the role **Operator**, **Change Password**, **Manage Operators** and **Log out**.

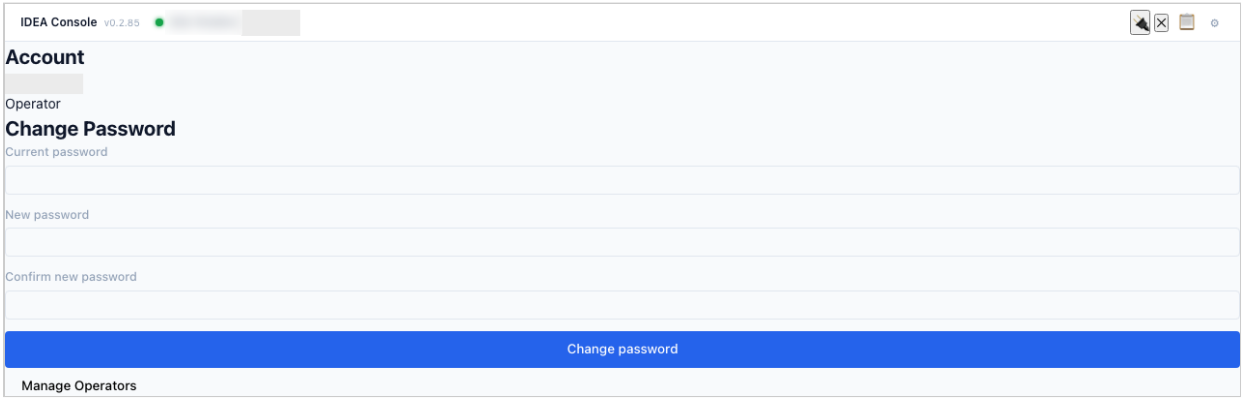


Figure 2. Account panel

3. Close the panel with ×. Use **Log out** in the same panel when you are done.

2.2 The Network view: the IDEA box and its disks

1. After login the left side shows **NETWORK**: **ALL APPS**, then the IDEA box with a green dot (online) and its disks. A box with a **red dot** is offline: switched off, or at another school.

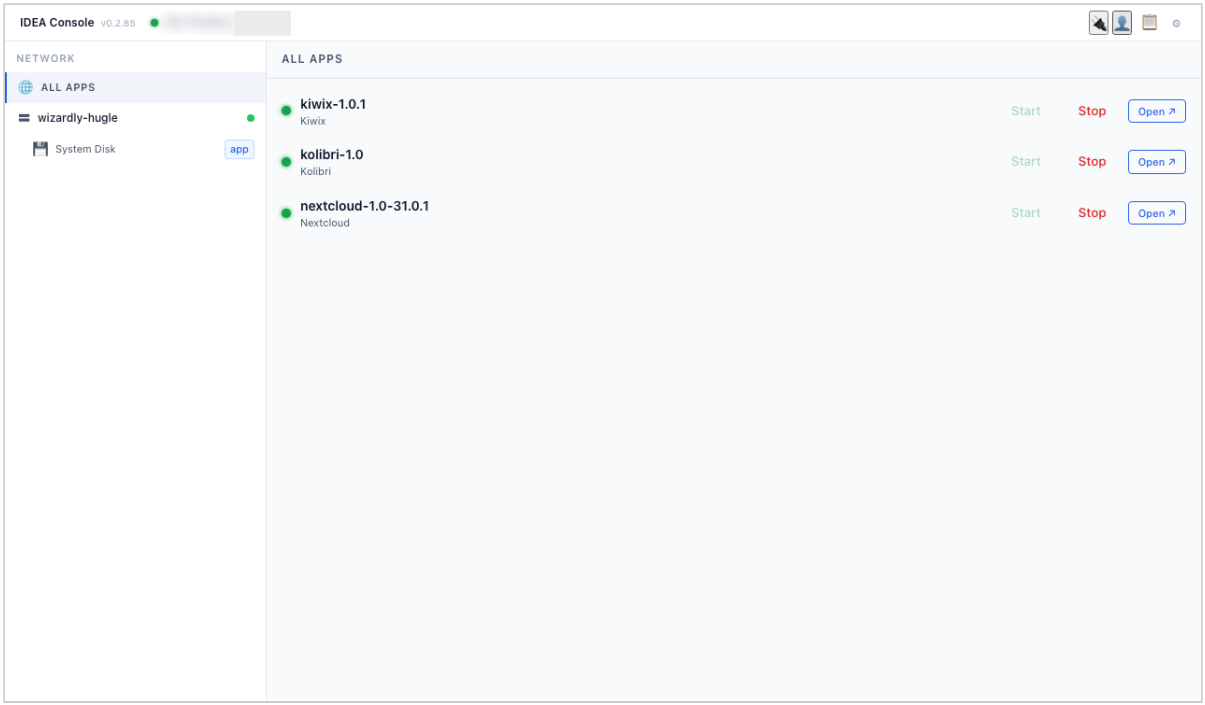


Figure 3. All apps

2. Click the IDEA box name to select it, then click a disk (here **System Disk**). Each disk shows a role badge: **app**, **backup**, **files** or **empty**. The right side lists the apps on that disk.

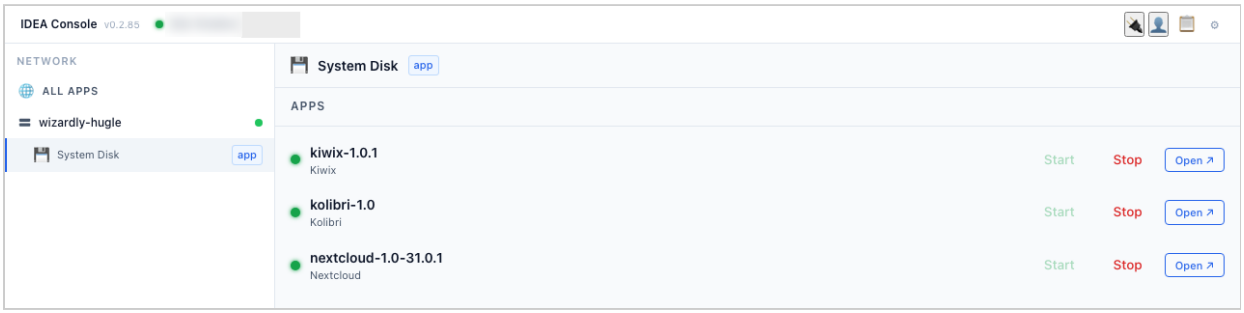


Figure 4. Disk view

2.3 App status

1. Each app row shows the app name and version, a coloured status dot, and the buttons **Start**, **Stop** and **Open ↗**.

2. Click the status dot (**Show details**) to see **STATUS** (for example **Running**), **VERSION**, **PORT**, **CREATED**, **LAST STARTED**, **LAST BACKUP** and the description.

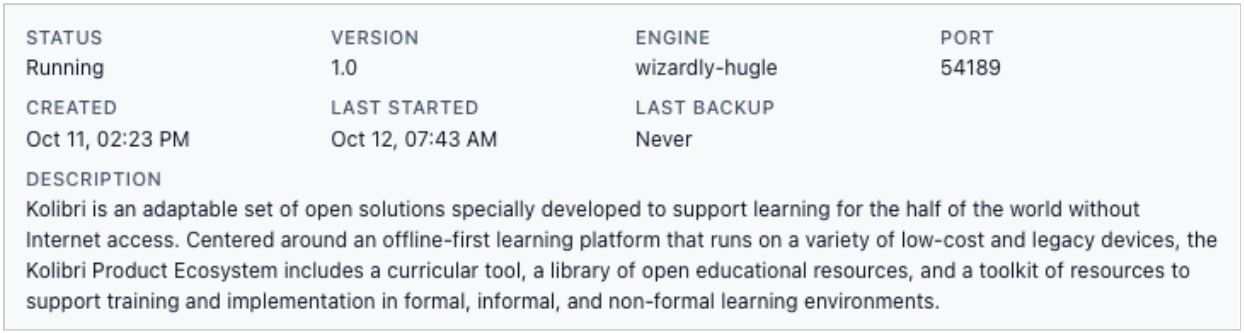


Figure 5. App details

3. To check that an app really works, click **Open ↗**: the app's page must load.
4. Only start or stop an app when your IDEA contact asks you to. Do not touch apps your IDEA contact did not mention.
5. **Known problem.** The apps of an IDEA box with a red dot still show as running, in **ALL APPS** and on the teachers' **Apps** page, but they won't open. There is no fix yet. Do not try to start or stop them: just tell the teachers to use the apps of your own box.

2.4 Command History

The clipboard icon (top right) lists recent Console commands with ✓ (done) or ✗ (failed) and how long ago. Read it out to your IDEA contact when you report a problem.

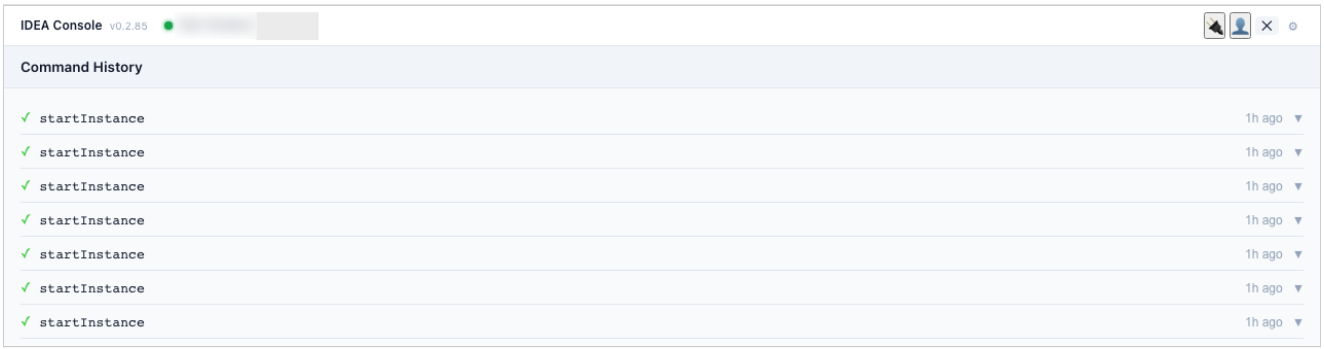


Figure 6. Command History

2.5 What the buttons do

- **Start** and **Stop** start or stop an app. A stopped app cannot be opened.
- **Install App** puts a new app on a disk.
- **Make this a Files Disk** turns an empty IDEA Disk into a Files Disk for Nextcloud (chapter 3).
- **Make this a Backup Disk** turns an empty IDEA Disk into a Backup Disk (chapter 3).
- **Restore** brings an app back from the Backup Disk onto a spare disk (chapter 3).
- **Erase this disk…** deletes everything on a disk. The System Disk has no Erase button.
- The small ↺ next to the IDEA box name is **Reboot**. It restarts the whole IDEA box, so all apps are away for a few minutes. Use it outside lesson time.

3. Disks: Files Disk and Backup Disk

3.1 A new disk (not yet an IDEA Disk)

1. A new disk appears under the IDEA box with its model and size and an **UNFORMATTED** badge.

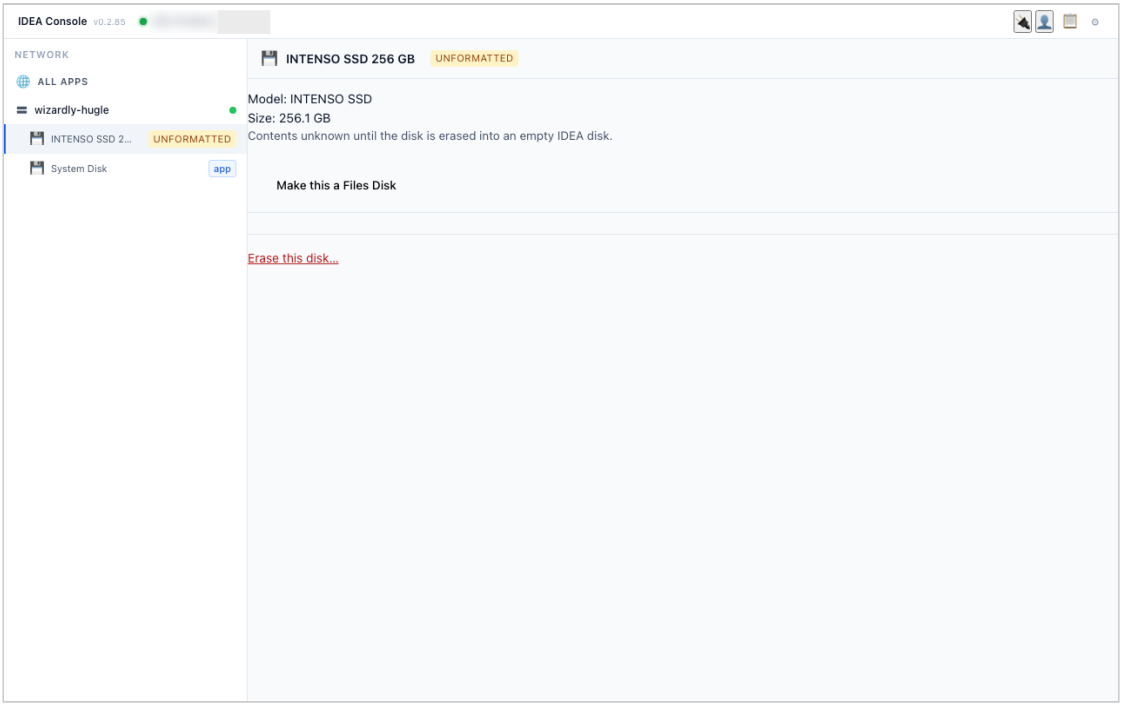


Figure 7. Unformatted disk

2. Its view shows the model, the size and two buttons: **Make this a Files Disk** and **Erase this disk…**.

USB sticks. A teacher's FAT32 or exFAT stick shows as **USB stick: (name) (read-only)**. Do not erase it: **Erase this disk…** turns the stick into an IDEA Disk and deletes the teacher's files, so only do it when asked.

3.2 Erase a disk

1. Click **Erase this disk…**. The confirmation shows the disk name, model, size and contents, and the red warning "This erases everything on the disk…".
2. Check the **size and contents**: several disks can have the same name (for example **IDEA Disk**). Make sure it is not the Backup Disk.

3. The box above the text field says **TYPE IDEA DISK TO CONFIRM** (the disk name in capitals). Type the name as it is written on the disk, for example `IDEA Disk`. The **Erase this disk** button stays grey until the name is right. Press **Cancel** if anything looks wrong.

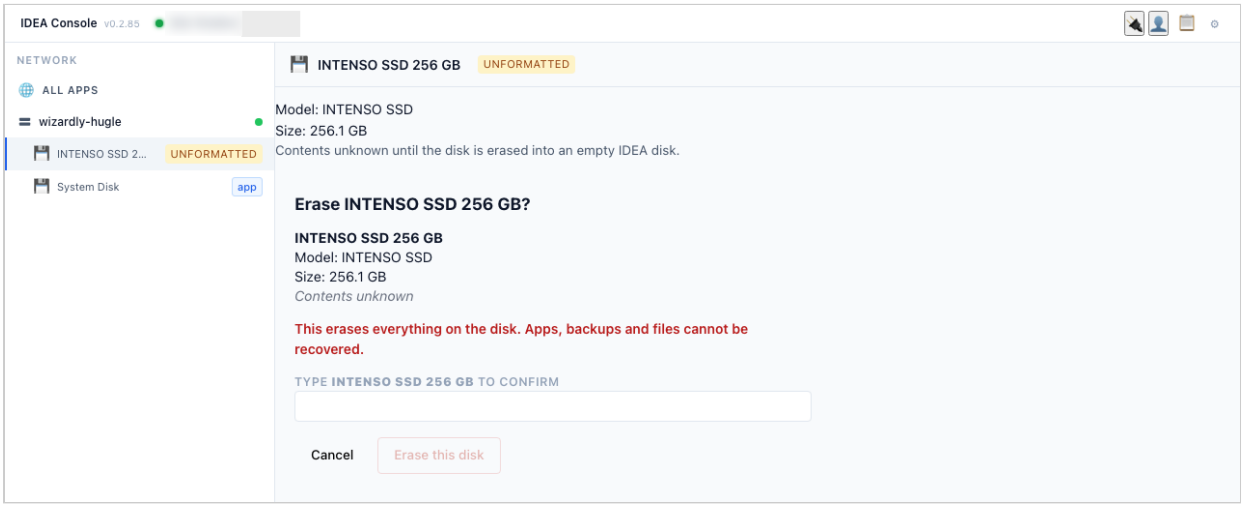


Figure 8. Erase confirmation

4. After the erase, the disk shows as an **IDEA Disk** with the **empty** badge.

> **A Files Disk that Nextcloud is using cannot be erased.** Ask your IDEA contact to stop Nextcloud first.

3.3 Make a Files Disk

1. Select the **IDEA Disk** marked **empty**. The view asks "What would you like to do with this disk?": **Make this a Files Disk**, **Make this a Backup Disk** or **Install App**.

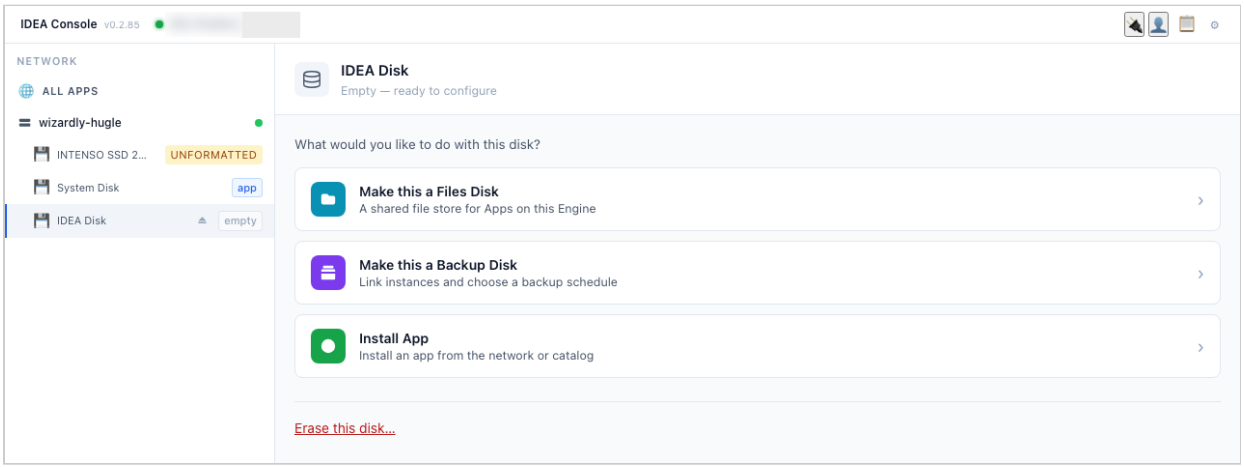


Figure 9. Empty IDEA Disk

2. Click **Make this a Files Disk**. The panel says "Nothing on the disk is erased", with **SHARE NAME** filled in as **School Files**. Click **Make this a Files Disk**.

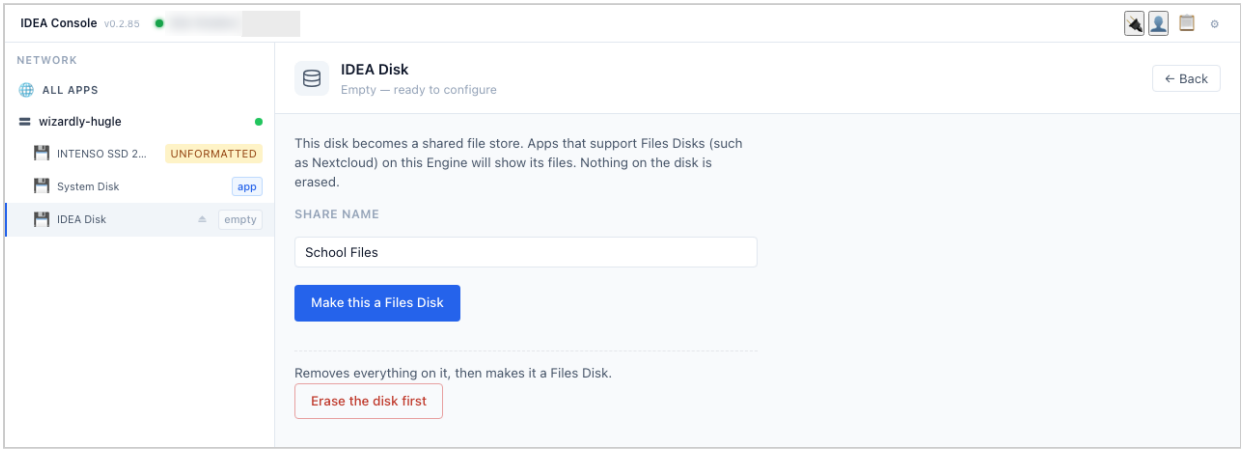


Figure 10. Make this a Files Disk

3. After a short wait the disk shows the **files** badge, "Share name: School Files", the free space, an **Eject** button and **Erase this disk...**. Once your IDEA contact has linked it, the view also says **Available in: Nextcloud**. Teachers then find it in Nextcloud (IDEA for Teachers: Tutorial, chapter 8).

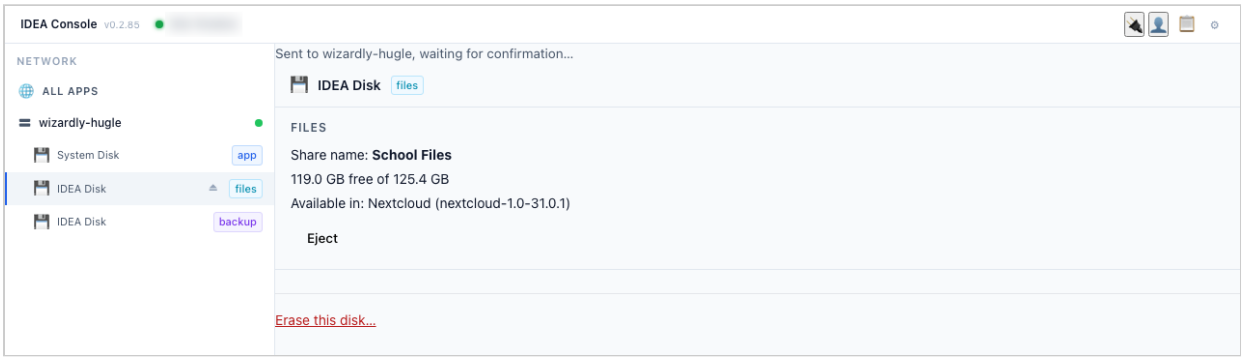


Figure 11. Files Disk available in Nextcloud

3.4 Eject a disk

An IDEA Disk has a small eject icon next to its name in the Network list and an **Eject** button in its disk view. The System Disk has none. Never eject or unplug a disk during a backup.

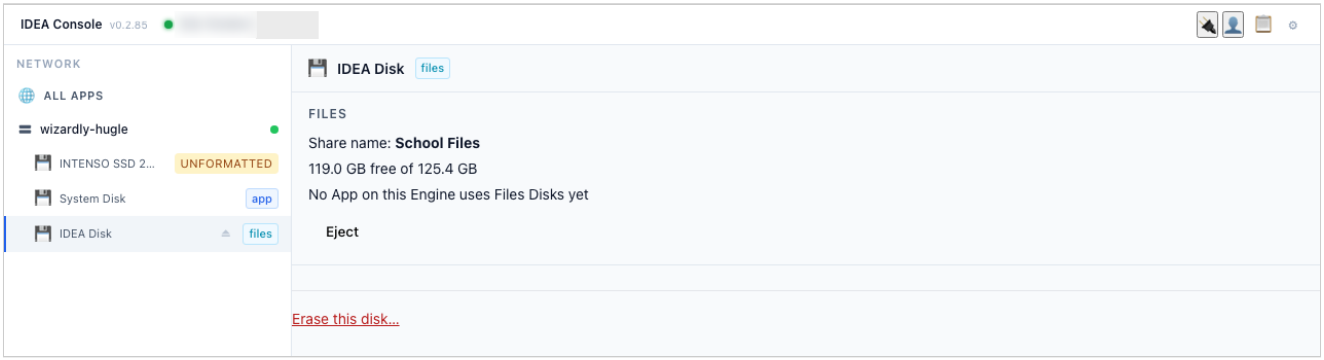


Figure 12. Eject button on a Files Disk

3.5 Make a Backup Disk

1. Select an **IDEA Disk** marked **empty** and click **Make this a Backup Disk**.

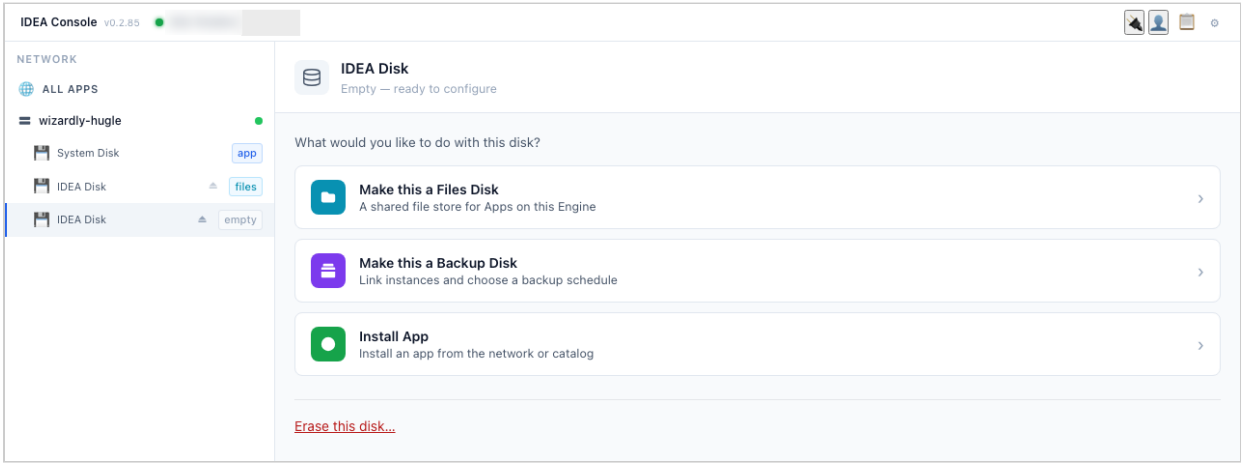


Figure 13. Empty disk options

2. Choose when backups run: **On demand** (only when you press **Back up**), **Immediate** (as soon as the disk is plugged in) or **Scheduled**. Your IDEA contact tells you which.

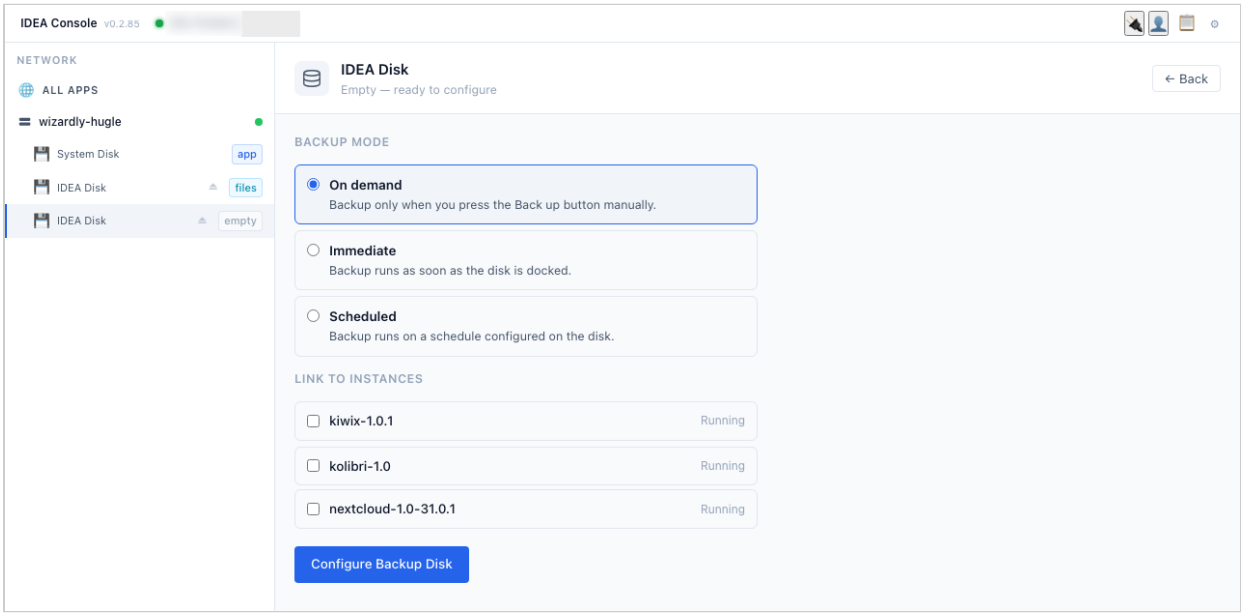


Figure 14. Backup mode

3. Under **LINK TO INSTANCES**, tick every app of this IDEA box (on the kit: all six). Backups are full: they include the Kolibri channels and Wikipedia, so a restore needs no internet.

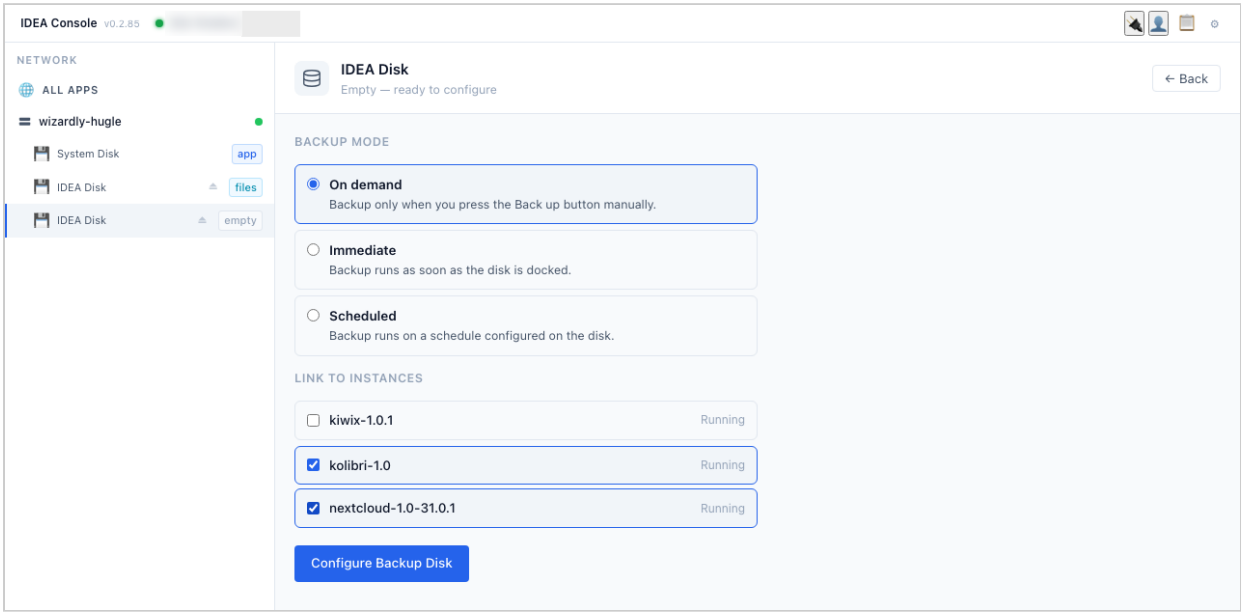


Figure 15. Linked apps

4. Click **Configure Backup Disk**. The disk gets the **backup** badge and a **Backups** list with each linked app.

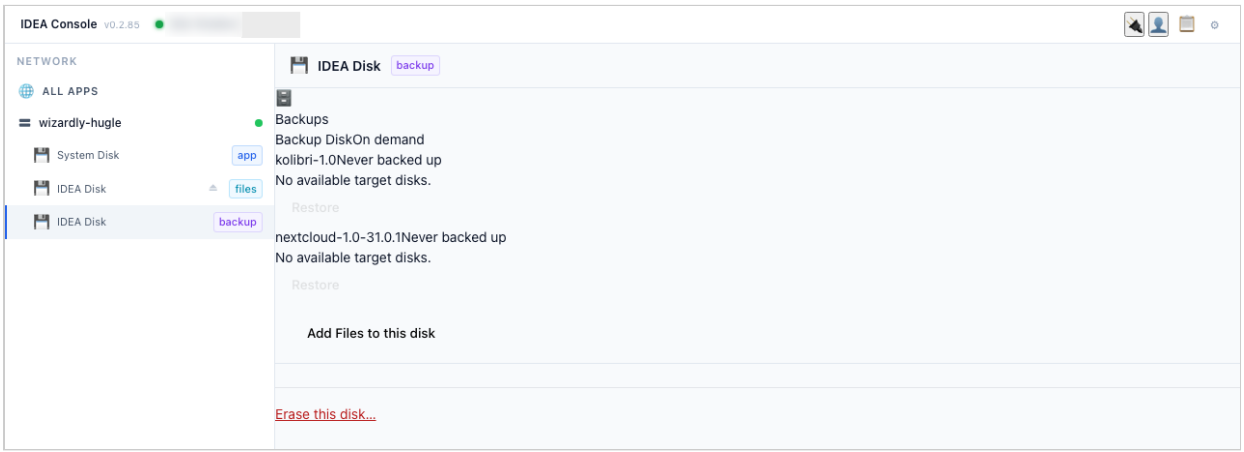


Figure 16. Backup Disk configured

3.6 Run a backup

1. In **ALL APPS**, each linked app row has a **Back up** button next to Start, Stop and Open. Click it for Kolibri. The row shows **Backing up…** with a progress bar, and the **OPERATIONS** box at the top shows the same job.

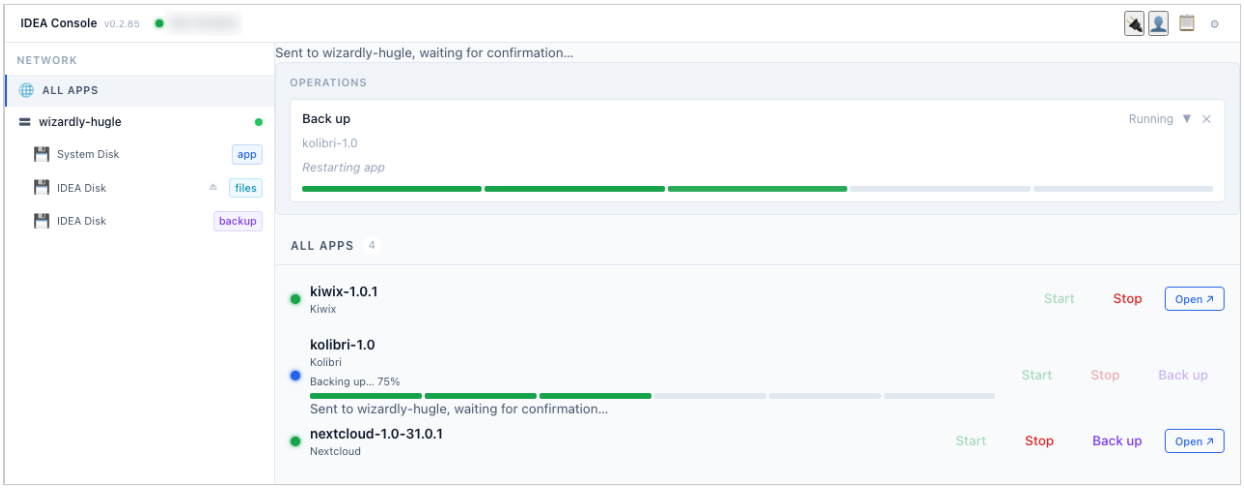


Figure 17. Back up clicked

2. The app is stopped while it is copied, then starts again by itself. When the job is finished the OPERATIONS box shows **Done**; close it with ×. A small app takes a few minutes, but the first backup of a big one takes much longer: on the kit about 37 minutes for Kolibri, an estimated 1 h 30 for the full Wikipedia and about 2 h 15 for all apps together. Do this outside lesson time.
3. When it is done, the Backup Disk's **Backups** list shows **Last backup** with the date and time for each app. Repeat for each other app.

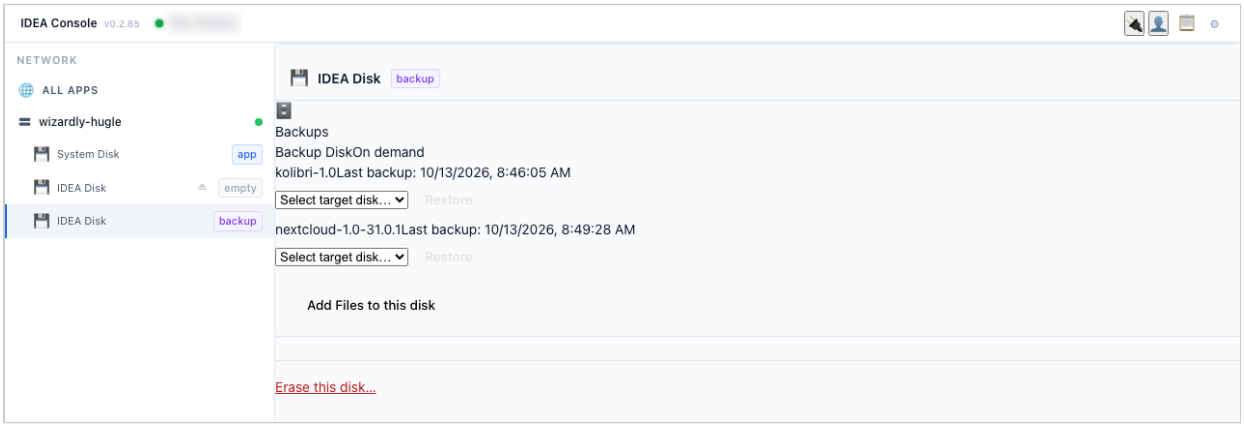


Figure 18. Last backup for both apps

3.7 Restore after a breakdown

You need a **spare disk** (an empty IDEA Disk) to restore onto.

1. Open the **Backup Disk**. Each app in the **Backups** list shows **Last backup**, a **Select target disk…** list and a **Restore** button (grey until a target is chosen).

2. In **Select target disk…** choose the spare disk. Only spare disks are offered here. Click **Restore**.

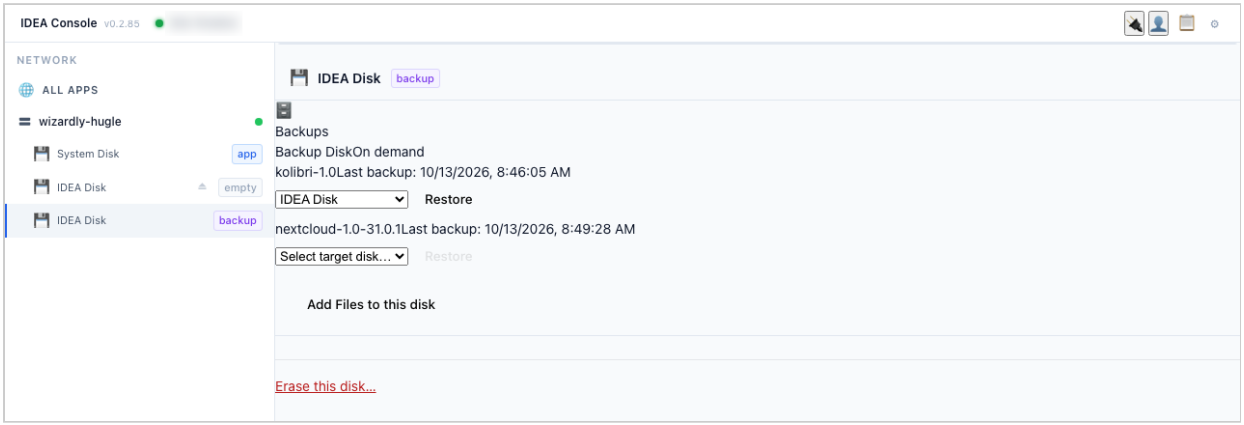


Figure 19. Target chosen

3. The Console asks: "Are you sure? This will overwrite **kolibri-1.0** on **IDEA Disk**." Click **Confirm Restore**.

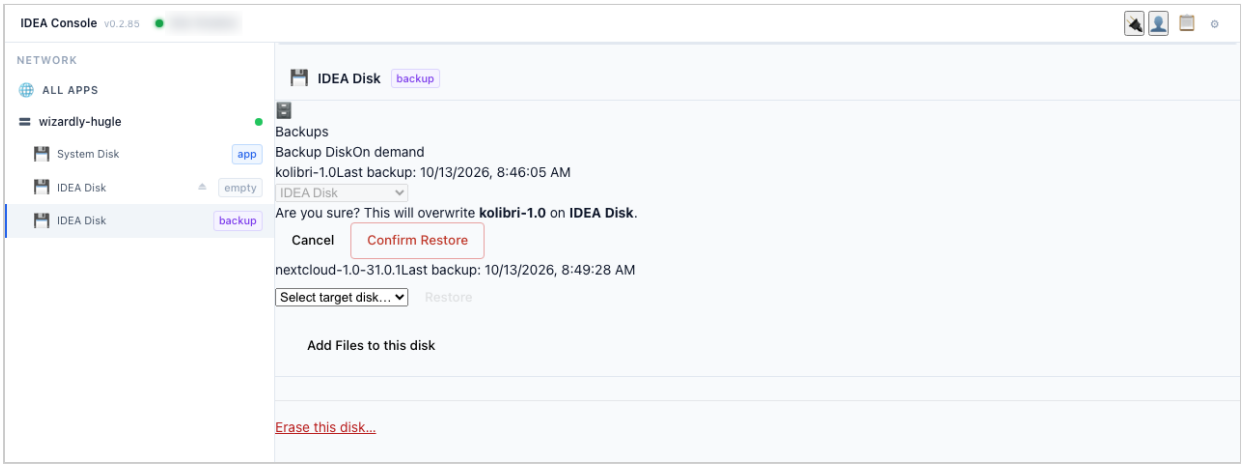


Figure 20. Confirm Restore

4. When the restore is finished, the spare disk shows the **app** badge and lists the restored app. A small app takes about a minute; Kolibri with its channels or Wikipedia takes much longer, so wait.

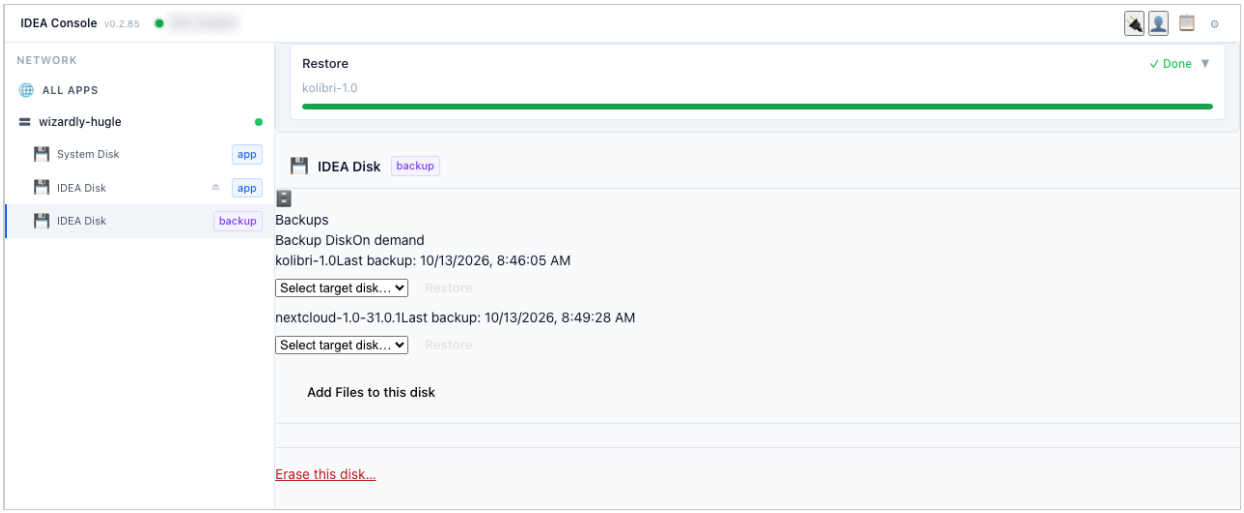


Figure 21. Restore done

5. The restore **moves** the app: it is no longer listed on its old disk, and the restored app is **stopped** (orange dot, **Start** button). Until you start it, the app cannot be opened. A restore is for a real breakdown, not for practice on a working school box.
6. Click **Start** on the restored app's row. After about 30 seconds it runs again and the row shows **Open ↗**.

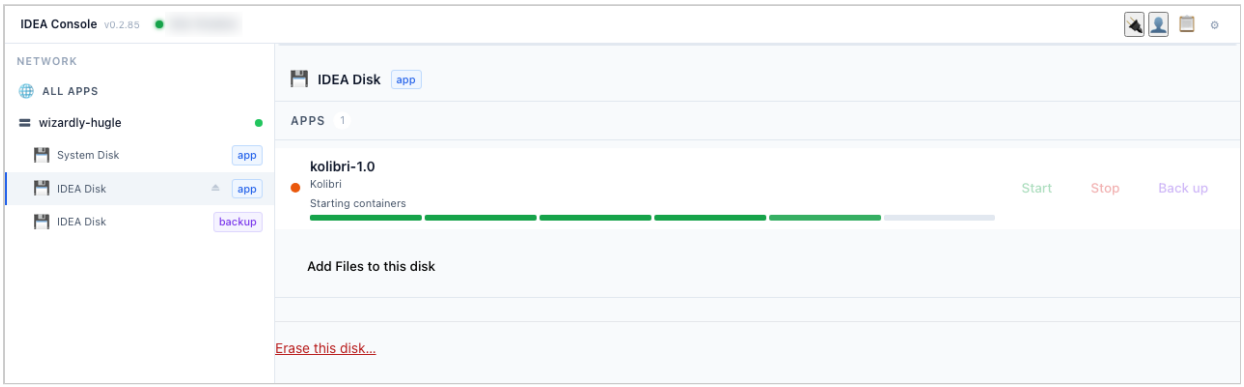


Figure 22. Start clicked

7. Open the app from its card on the **Apps** page (as in IDEA for Teachers: Tutorial, 2.3) and sign in as the teacher (coach).

8. Check that the data is back: classes, learners, lessons, quizzes and reports, as at the last backup.

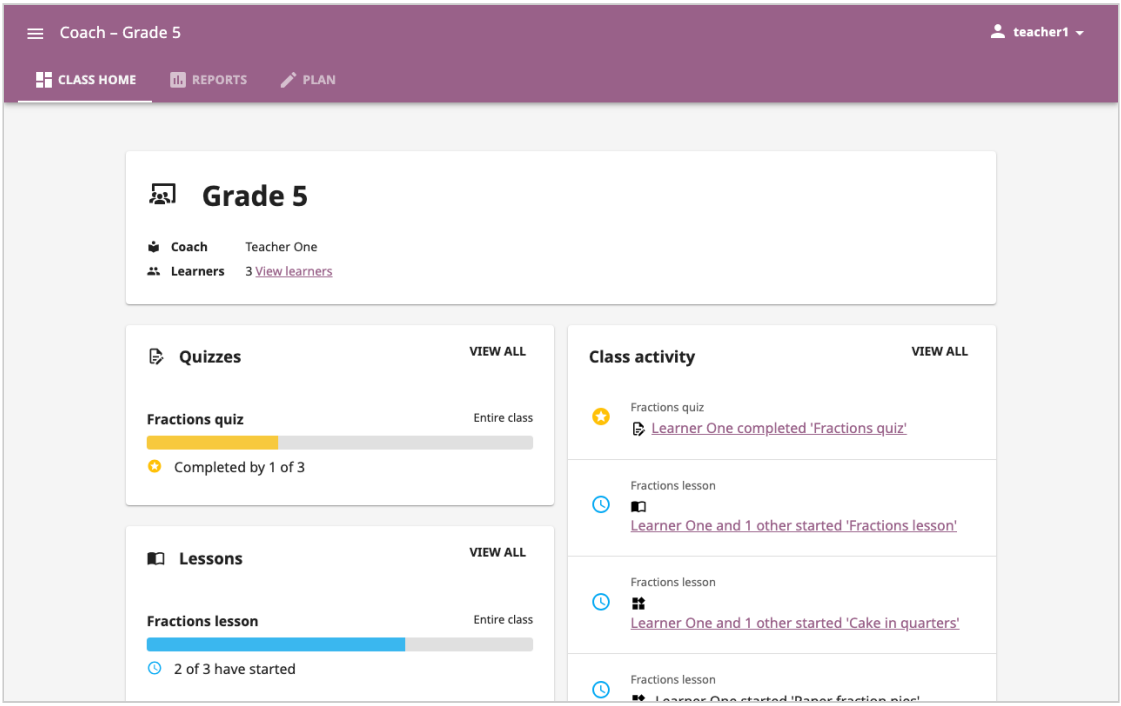


Figure 23. Class after restore

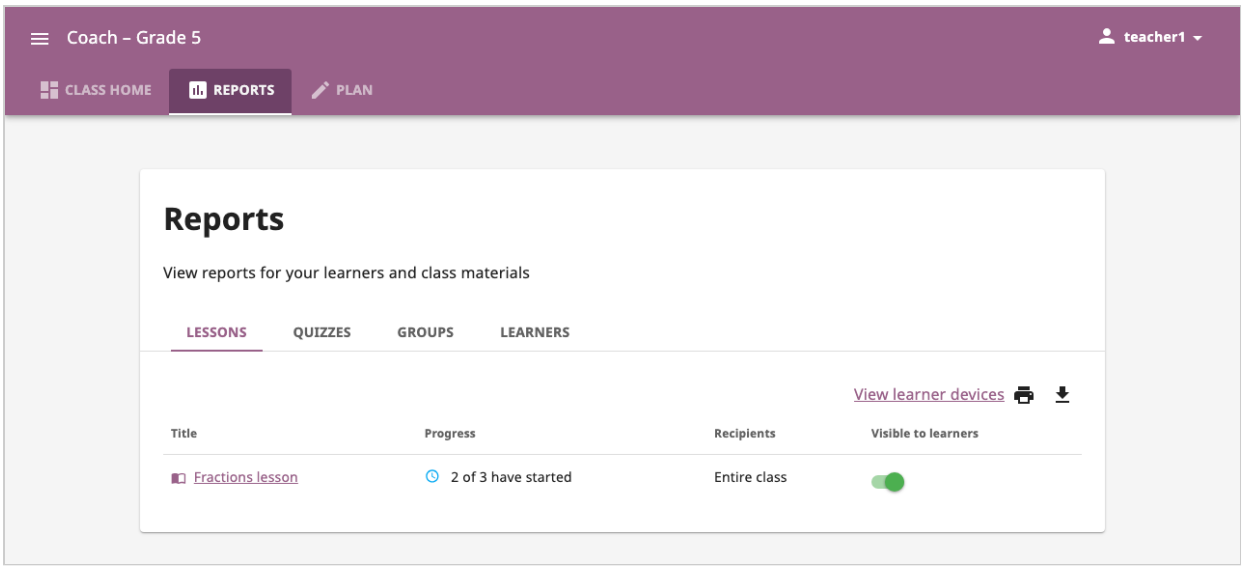


Figure 24. Reports after restore

9. Later, your IDEA contact can move the app back to the System Disk and prepare the spare disk again.

Important. Anything done after the last backup is lost. Make a new backup of the restored app as soon as it runs.

4. Student computers: reset a card from the Console

Every student computer (Raspberry Pi 400 or Pi 500) starts from a **microSD card** with the same IDEA student image. When a computer goes wrong (it no longer starts, the screen looks different, the card is full or a student changed settings), you do not repair it. You **reset the card**: the IDEA box writes a fresh copy of the student image onto it. This takes about 5 to 10 minutes.

Good to know:

- **Resetting wipes the card.** Anything saved on the computer itself (the local Documents folder) is gone. Students' work in **Nextcloud** stays: that is why students save there (IDEA for Teachers: Tutorial, chapter 6).
- Every card is the same, so a card works in **any** student computer, Pi 400 or Pi 500.
- The student computers **do not update themselves**. New software comes only with a new, tested image version, which you get by resetting the card.
- **A restart already undoes most problems.** Every restart puts the computer back as its card was written (files and settings saved on the computer are gone). Try a restart first; reset the card only when that doesn't help.
- You need a **USB card reader** plugged into the IDEA box. Any card reader works.

4.1 Before you start

1. Switch the student computer off and take the microSD card out. Note the computer's name from its sticker (for example `idea-pc-3f2a`).
2. Put the card into the card reader, and plug the card reader into a free USB port of the IDEA box.
3. Open the Console and sign in with **Operator Login** (chapter 2).

4.2 Find the card

1. Click the IDEA box in the list on the left. Scroll down to **CLIENT CARDS**. It shows the client image the box will write, and every card in a card reader: the reader's name, the card's size and what is on it now.

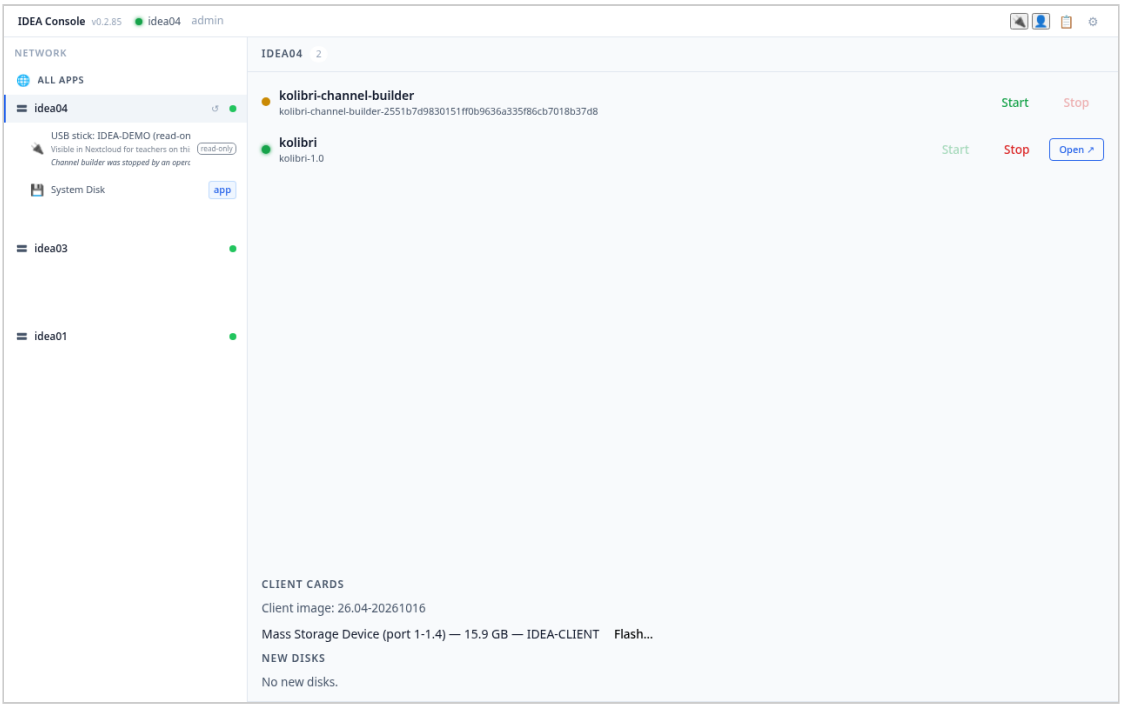


Figure 25. Client cards: one card in a reader, with Flash...

2. Check that the **size** matches your card. If the card is not listed, see 13.5.

> **Never pick an IDEA Disk.** The Files Disk, the Backup Disk and app disks are never offered here, but always check the name and size before you continue.

4.3 Reset (flash) the card

1. The card belongs to the IDEA box you flash it from: its **IDEA** shortcut tries that box first. Flash it from the IDEA box of the classroom it will be used in.
2. Click **Flash...** on the card's row.
3. Read the panel **FLASH CLIENT CARD**. It repeats the card, its size, what is on it now and the image. It says in red that everything on the card will be wiped. If anything looks wrong, click **Back**.

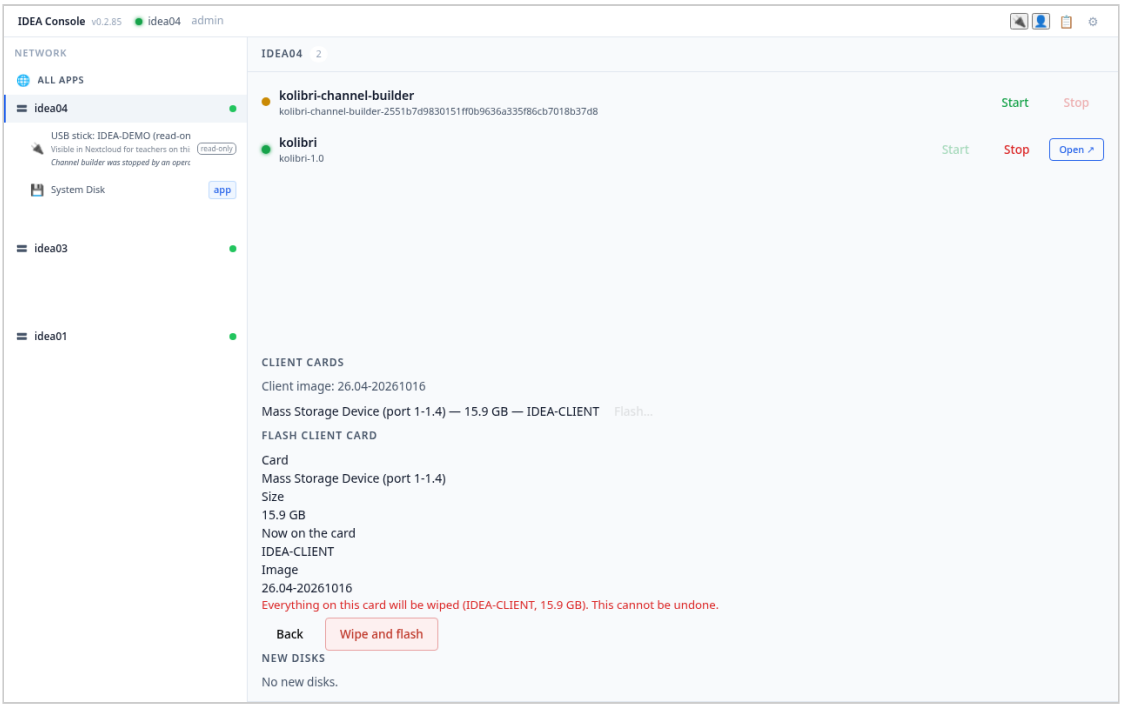


Figure 26. Flash client card: check the card, then Wipe and flash

4. Click **Wipe and flash**. The Console shows the progress in steps, for example **Step 3 of 6: Writing**.

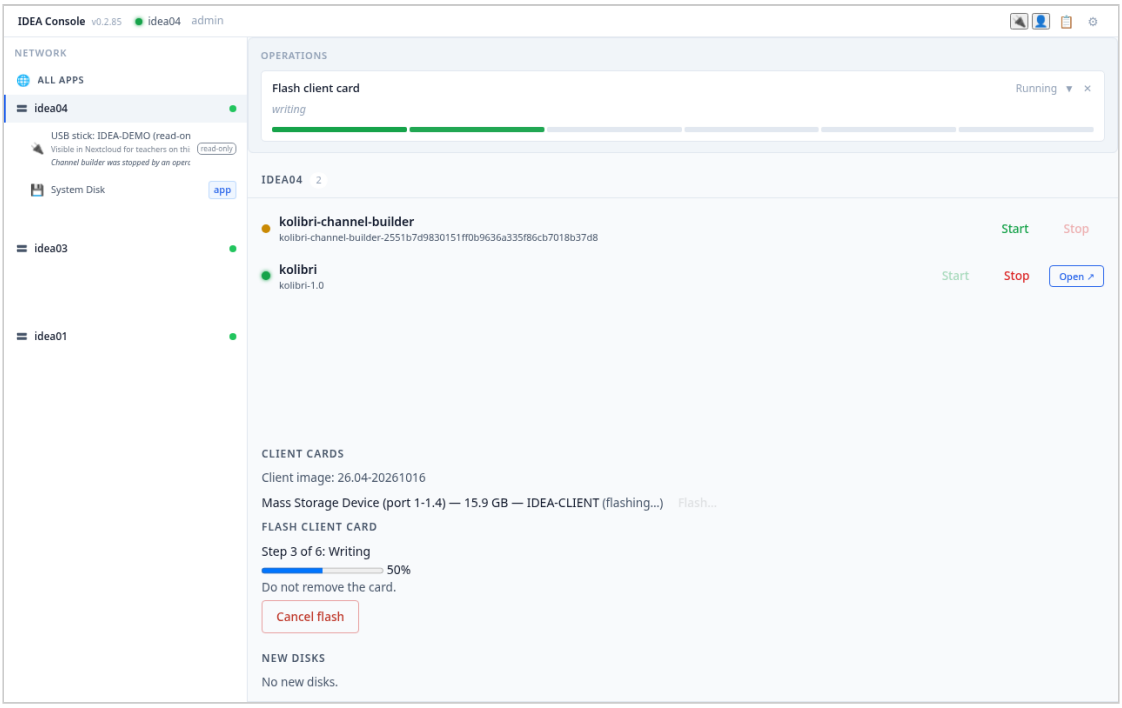


Figure 27. Flash in progress

5. Wait for **Done – safe to remove**. Do not pull the card reader out before that. Then click **Close**.

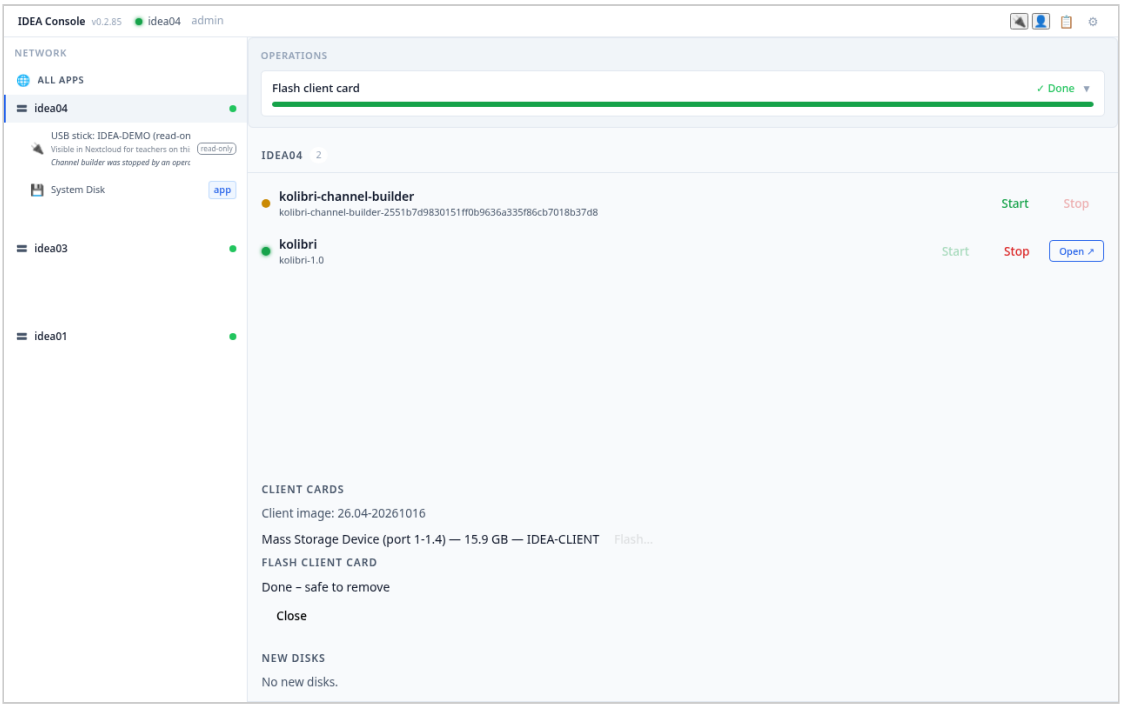


Figure 28. Done – safe to remove

6. Take the card out of the reader and put it back into the student computer. Switch the computer on.

4.4 After the reset

1. The first start takes a few minutes longer: the card sets itself up and names the computer after its Pi (the name on the sticker).
2. The computer opens the student desktop by itself, with no password. The desktop has one shortcut, **IDEA**.

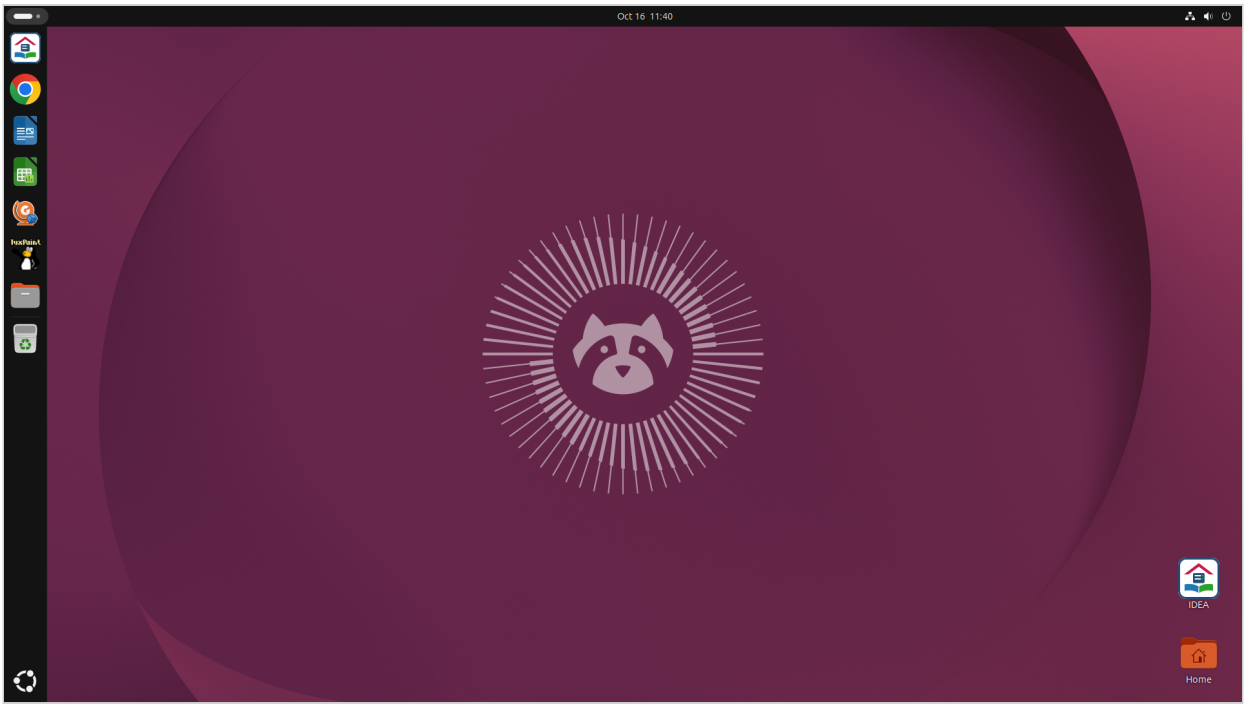


Figure 29. The student desktop after the first start, with the IDEA shortcut

3. Double-click **IDEA**. Chrome opens the start page, finds the IDEA box and opens the Console (IDEA for Teachers: Tutorial, chapter 2). Check that **Open** on the Kolibri card works.

4.5 Questions

Question	Answer
The card is not listed.	Check that the card is pushed fully into the reader and that the reader is plugged into **this** IDEA box. Try another USB port.
The Flash button is not there or the Console refuses.	The IDEA box never writes to an IDEA Disk, the system disk, or a disk that is in use. Check that you picked the card and not another disk. Otherwise ask your IDEA contact.
The flash stopped with an error.	Flash again. If it fails again, the card is probably worn out: use a new card.
The computer does not start after the reset.	Try the card in another student computer. If it starts there, the computer may need a bootloader update: ask your IDEA contact.
Can I install an extra program on one computer?	No. It would be lost at the next reset. Ask your IDEA contact to add it to the next image version.
The IDEA shortcut says No IDEA server found.	Check the network connection. Type the name on the IDEA box's sticker in the field on the start page (IDEA for Teachers: Tutorial, chapter 2). If it keeps happening, the card may belong to an IDEA box that has been switched off or moved: flash it again from the right box.
Can my IDEA contact do this remotely?	Yes. Put the card in the reader and tell them. They press Flash from the Console, and you put the card back when the Console says it is done.